Two stabilizers and elementary matrices in dimension three

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We reprove the rank-four construction of Leemans and Vandenschrick [1, Theorem 1.1]. Its matrices and duality are retained exactly. The generation argument below uses elementary matrices; it does not require a classification of maximal subgroups. We retain the classical rotation-group correspondence and its rank-four intersection criterion [2], as already stated explicitly in the project's common theory for projective markings. We also retain the classical description of automorphisms of $\mathrm{PSL}_3(q)$ as projective semilinear maps and their compositions with inverse transpose, for $q \geq 5$.

Let $F = \mathbb{F}_q$, where $q \geq 5$, and let x generate $F^\times$. Put

$$a = \begin{pmatrix} x & 1 & 0 \\ 0 & 1+x^{-1} & -x^{-1} \\ 0 & 1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} -x^{-1} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & x & 0 \end{pmatrix}, \quad c = \begin{pmatrix} x & 0 & 0 \\ x-x^{-1} & 1 & 0 \\ x-x^{-1} & 0 & x^{-1} \end{pmatrix}.$$

All have determinant one. We use the same letters for their projective images. Set $d = \gcd(3, q-1)$.

Theorem 1. *The ordered triple (a, b, c) generates $\mathrm{PSL}_3(q)$ and defines a self-dual chiral polytope of type*

$$[q-1,\ 2(q-1)/d,\ q-1].$$

Its two end subgroups are the full set stabilizers of two distinct lines and two distinct points, respectively. Each has order $2q^2(q-1)^2/d$.

1 The full stabilizers

For $u, v \in F$ and $r \in F^\times$, write

$$K(u,v) = I + uE_{21} + vE_{31}, \qquad h(r) = \mathrm{diag}(r^{-2}, r, r).$$

Thus $K(u,v)K(u',v') = K(u+u', v+v')$ and $b^{2m} = h(x^m)$. Direct multiplication gives, for $r, s \in F^\times$,

$$h(r)c^{-1}h(s/r)ch(s^{-1}) = K(C/x, C), \quad C = (r^3 - s^3)(1-x^2),\ c^{-1}K(u,v)c \ = K(xu, x^2v). \quad (1)$$

The factor x^{-1} in the first coordinate of the first identity is essential. The corresponding displayed word on source page 6 omits it.

The $\mathbb{F}_p$-span V of all $r^3 - s^3$ is stable under multiplication by x^3 and contains $x^3 - 1 \neq 0$. Moreover $\mathbb{F}_p(x^3) = F$: otherwise x^3 would belong to a proper subfield of size at most $\sqrt{q}$, whereas

$$|x^3| = (q-1)/\gcd(3, q-1) \ \geq \ (q-1)/3 \ > \ \sqrt{q} - 1.$$

The strict inequality holds for $q > 4$. Hence V is a nonzero vector space over F, so $V = F$. Since $1 - x^2 \neq 0$, the first identity and addition in K give every $K(t/x, t)$. Conjugation by c

gives every $K(t, x^2 t)$. The vectors $(x^{-1}, 1)$ and $(1, x^2)$ are independent, with determinant $x - 1$. It follows that

$$K(F, F) \;\leq\; \langle b, c \rangle.$$

Let $p = [0, 1, 0]^t$, $p' = [0, 0, 1]^t$. Their full set stabilizer in $\mathrm{SL}_3(F)$ consists of the matrices

$$\begin{pmatrix} \alpha & 0 & 0 \\ u & \beta & 0 \\ v & 0 & \gamma \end{pmatrix}, \quad \alpha\beta\gamma = 1, \qquad \text{and their products with } b.$$

The subgroup fixing both points is $K(F, F)$ times the determinant-one diagonal torus. Modulo K, the diagonals of $b^{2m} c^n$ have exponent vectors

$$(-2m + n,\; m,\; m - n).$$

Every triple of exponents summing to zero occurs: for exponents (A, B, C) take $m = B$ and $n = A + 2B$. Thus b, c generate the full set stabilizer, even before taking projective images. Dividing its order by the scalar center gives $2q^2(q-1)^2/d$.

Put

$$D = \begin{pmatrix} (x - x^{-1})^{-1} & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & x \end{pmatrix}, \qquad \phi(g) = D(g^{-1})^t D^{-1}.$$

The matrix D is symmetric and invertible. Consequently $\phi^2 = 1$, and direct multiplication gives

$$\phi(a) = c^{-1}, \qquad \phi(b) = b^{-1}, \qquad \phi(c) = a^{-1}.$$

It interchanges $\langle b, c \rangle$ and $\langle a, b \rangle$. The latter is therefore the full stabilizer of the unordered pair of lines

$$\ell : y - z = 0, \qquad \ell' : y - x^{-1} z = 0.$$

One can also identify these lines directly from the actions of a, b; a fixes them and b exchanges them.

2 Generation and intersection

Let $\widetilde{G} = \langle a, b, c \rangle \leq \mathrm{SL}_3(F)$. The preceding argument gives all $I + tE_{21}$ and $I + tE_{31}$ in $\widetilde{G}$. Since ϕ preserves $\widetilde{G}$, it also gives the images of all these matrices under ϕ. The block diagonal shape of D shows that these images range over all

$$I + uE_{12} + vE_{13}.$$

Indeed inverse transpose changes $K(u, v)$ into $I - uE_{12} - vE_{13}$, and conjugation by D applies an invertible linear map to (u, v). Commutators now give the remaining root groups:

$$[I + sE_{21}, I + tE_{13}] = I + stE_{23}, \qquad [I + sE_{31}, I + tE_{12}] = I + stE_{32},$$

where $[g, h] = ghg^{-1}h^{-1}$. These six elementary root groups generate $\mathrm{SL}_3(F)$ by row elimination. This proves generation without any overgroup classification.

Write $H = \langle b, c \rangle$ and $A = \langle a, b \rangle$ in the projective group. An element of $A \cap H$ preserves both unordered pairs $\{p, p'\}$ and $\{\ell, \ell'\}$. It fixes $e = [1, 0, 0]^t = \ell \cap \ell'$, and hence has a diagonal or a monomial representative

$$\mathrm{diag}(\alpha, \beta, \gamma), \qquad \begin{pmatrix} \alpha & 0 & 0 \\ 0 & 0 & \beta \\ 0 & \gamma & 0 \end{pmatrix}.$$

On the line pp', the two distinguished intersections with ℓ, ℓ' have ratios $z/y = 1, x$. A diagonal representative preserves this pair only when $\gamma/\beta = 1$, since $x^2 \neq 1$. The determinant then makes it $h(\beta)$, a power of b^2. A monomial representative sends z/y to $\gamma/(\beta(z/y))$; pair preservation forces $\gamma/\beta = x$. Its determinant makes it $h(\beta)b$, again a power of b. Thus $A \cap H = \langle b \rangle$.

Every power of c fixes p, p' separately. An element shared with $\langle b \rangle$ must therefore be an even power of b, acting as a scalar on pp'. But c^n acts there with eigenvalue ratio x^n, so it is shared only if $q - 1$ divides n. The eigenvalues of c are the distinct elements $x, 1, x^{-1}$; hence its projective order is $q - 1$ and this shared element is the identity. Applying ϕ gives the other adjacent cyclic intersection. We have proved

$$\langle a \rangle \cap \langle b \rangle = 1, \quad \langle b \rangle \cap \langle c \rangle = 1, \quad A \cap H = \langle b \rangle.$$

Direct multiplication gives $(ab)^2 = (bc)^2 = (abc)^2 = I$; each product is nonscalar. The retained rank-four criterion now proves the full rotation intersection property.

3 Excluding reversal

Suppose an automorphism α of $\mathrm{PSL}_3(F)$ sends a to a^{-1} and fixes ab and abc. Then it fixes $c = (ab)^{-1}(abc)$ and preserves $A = \langle a, ab \rangle$.

The group H fixes no projective point. Indeed its subgroup $K(F, F)$ fixes precisely the points on pp'; the full diagonal torus has only p, p' as common fixed points there, and b exchanges them. Dually, A fixes no line, although it fixes e. A graph automorphism followed by a semilinear collineation would take A to a group fixing a line. It therefore cannot preserve A. By the retained automorphism theorem, α is induced by a semilinear collineation with field part $t \mapsto t^{p^i}$, $0 \leq i < [F : \mathbb{F}_p]$.

The axes of ab and abc are respectively

$$z = 0, \qquad (1 - x)y_1 = z,$$

where y_1 denotes the first homogeneous coordinate. These follow, in both characteristics, from their displayed matrices; their axes are uniquely determined by the involutions. Their intersection is p, so the collineation fixes p. It also permutes the three eigenpoints of a, namely

$$[1, 0, 0]^t, \quad [1, 1 - x, 1 - x]^t, \quad [1, x^{-1} - x, 1 - x^2]^t.$$

Only the first lies on the invariant line $z = 0$, so it fixes e. The linear part P consequently has the form

$$P = \begin{pmatrix} u & 0 & w \\ 0 & v & t \\ 0 & 0 & s \end{pmatrix}.$$

Projective centralization of c means $Pc^{(p^i)}P^{-1} = \mu c$ for a scalar μ. Applying this equation to p, whose eigenvalue is one on both sides, gives $\mu = 1$. The other two eigenvalues then imply $x^{p^i} = x$ or x^{-1}. The second option is impossible: for $0 \leq i < [F : \mathbb{F}_p]$ and $q \geq 5$, $p^i + 1 < q - 1$. The first option implies $i = 0$, since x is primitive. The three eigenpoints of c must therefore be individually fixed. They are p, p' and $[x, 1 + x, x]^t$. Fixing p' gives $w = t = 0$; fixing the last point gives $u = v = s$, as $x(1 + x) \neq 0$. Thus P is scalar and α is the identity. This contradicts $\alpha(a) = a^{-1}$, because a has order $q - 1 > 2$. The marking is therefore chiral by the retained correspondence.

Finally, a, c both have order $q - 1$. Odd powers of b interchange p, p' and cannot be scalar; $b^{2m} = h(x^m)$ is scalar precisely when $x^{3m} = 1$. Hence $|b| = 2(q - 1)/d$. The automorphism ϕ sends the marking to its dual (c^{-1}, b^{-1}, a^{-1}), proving self-duality and the claimed type.

4 Attribution and scope

The matrices, stabilizer descriptions, duality and rank-four existence result are prior work of the source authors. We supply the finite-field span argument, the full elementary generation proof and the detailed projective scalar check in the reversal argument. The source's displayed unipotent word needs the factor x^{-1} noted above. Its next-page transitivity paragraph also interchanges the two named end subgroups; the elementary generation argument avoids that paragraph entirely. These corrections do not change Theorem 1.1.

This component treats Theorem 1.1 and its proof, not the source's rank-five construction, subgroup classifications or global rank obstruction. It gives no new answer to an open question. The automorphism theorem and the polytope correspondence remain explicit classical premises; finite examples are controls on the formulas, not the proof of the all-field statement.

References

[1] D. Leemans and A. Vandenschrick, *On chiral polytopes having a group* $\mathrm{PSL}(3, q)$ *as automorphism group*, fixed 25-page author edition, 2022, Theorem 1.1 and Section 3.

[2] E. Schulte and A. I. Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS 4 (1991), 493–516; rotation correspondence and rank-four intersection criterion.

[3] Human Compression project, *Common theory for projective markings and rotations*, Topic E, version 3, Section 2; retained rotation correspondence.